



WEXFORD WATER TECHNOLOGIES, LLC, DISPOSAL WELL

Wexford County, Michigan | March 2018

Overview

The Michigan Department of Environmental Quality (DEQ), Oil, Gas, and Minerals Division (OGMD), has received an application to drill and operate a Part 625 non-hazardous waste disposal well in Wexford County. Wexford Water Technologies, LLC, is the applicant for the proposed Wexford Water Technologies Well No. 1 (WWT 1). This proposed well will dispose of non-hazardous landfill leachate from the Wexford County Landfill. The entire application is available through the DEQ Web site, www.michigan.gov/deqland (mineral wells).

Landfill leachate is the liquid generated within a landfill. It is created from moisture contained within the waste itself, and the precipitation that falls on the landfill, which moves through the waste. Modern landfills are constructed with liners to collect leachate at the bottom of the landfill. Leachate is primarily water, with low concentrations of non-hazardous compounds. The pH of the wastewater is generally neutral, and is not considered ignitable, reactive, or corrosive. The wastewater will be tested regularly to monitor for various parameters and assure that it remains classified as non-hazardous.

Disposal wells of this type are permitted by both the OGMD and the U.S. Environmental Protection Agency (U.S. EPA), Underground Injection Control (UIC) Program. According to the U.S. EPA classification, this proposed well is considered a Class I disposal well. Class I disposal wells inject industrial and municipal wastewaters into deep, confined formations far below and isolated from freshwater zones. There are 23 active Class I disposal wells in Michigan and approximately 800 active Class I disposal wells in the United States.

Class I disposal wells are designed and constructed to allow for disposal of wastewater while preventing the movement of disposed wastewaters into (shallower) freshwater zones. Class I wells that meet design and operating requirements are very low risk. There are redundant protections and strict guidelines related to well location, design, construction, operation, monitoring, testing, reporting, and closure. The safeguards in place ensure protection to freshwater, the environment, and human health.

Wexford Water Technologies, LLC, proposes to dispose non-hazardous landfill leachate into deep, confined, porous rock formations between 3,100 and 6,000 feet below the ground surface. The disposal rock formations naturally contain high levels of salt. The base of the lowest known freshwater aquifer is approximately 900 feet below the ground surface. The proposed WWT 1 disposal well is designed to isolate the disposal zone from the freshwater zones with multiple strings of casing (steel pipes) cemented to the surface. The disposal is through steel tubing within the casing strings. There are several confining rock formations and over 2,000 feet of vertical separation between the disposal zone and the freshwater zone. Under no circumstances would a disposal well be approved or allowed to operate if there were indications that the disposed fluids could migrate into freshwater zones.

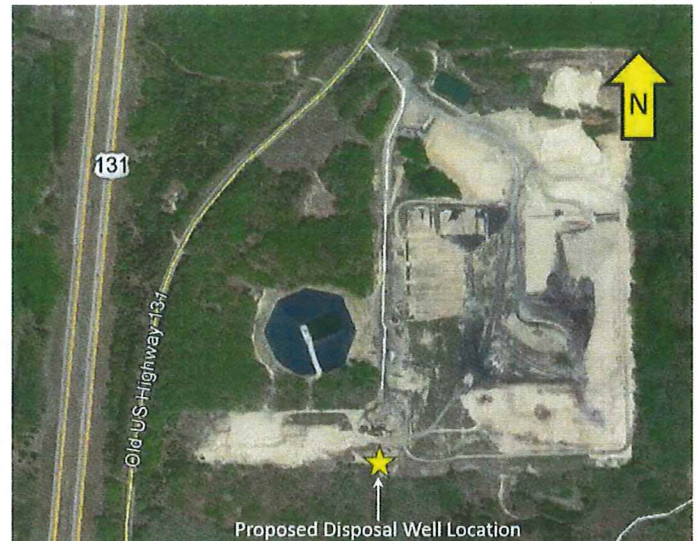


Figure 1. The proposed disposal well location is on the south side of the Wexford County Landfill property. (Aerial imagery source: Google Earth)

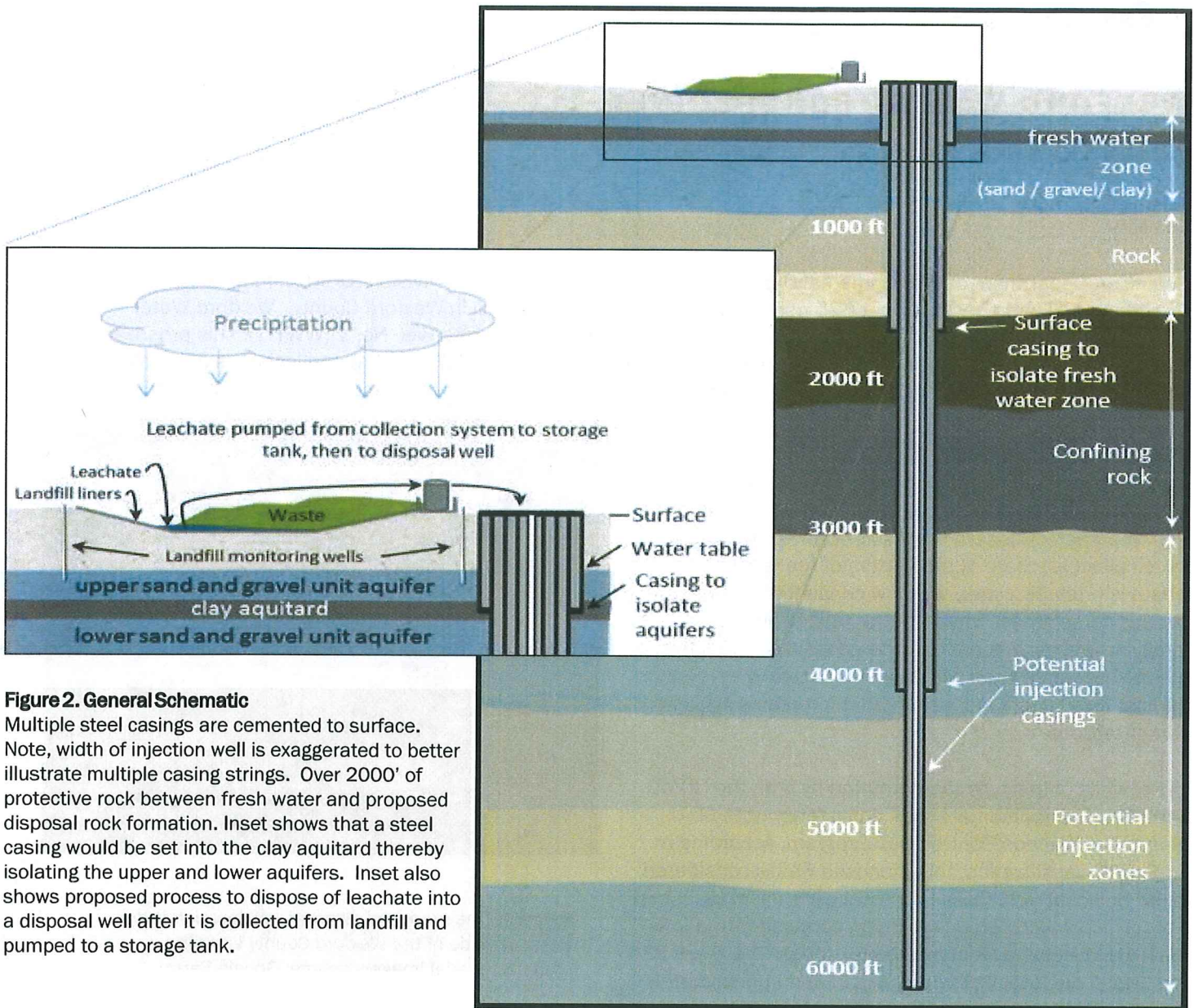


Figure 2. General Schematic

Multiple steel casings are cemented to surface. Note, width of injection well is exaggerated to better illustrate multiple casing strings. Over 2000' of protective rock between fresh water and proposed disposal rock formation. Inset shows that a steel casing would be set into the clay aquitard thereby isolating the upper and lower aquifers. Inset also shows proposed process to dispose of leachate into a disposal well after it is collected from landfill and pumped to a storage tank.

DEQ OGMD formal public comment period closes at 5pm March 28, 2018.

Comments may be submitted via email to:

DEQ-OGMDpubliccomments@michigan.gov

or mailed to:

MDEQ-OGMD,

Re: Comments WWT 1

PO Box 30256,

Lansing, MI 48909

Additional Information

Web Sites:

DEQ: www.michigan.gov/deqland (mineral wells)

U.S. EPA Underground Injection Control: www.epa.gov/uic

Primary Contact: DEQ, Oil, Gas, and Minerals Division. 800-662-9278 or 517-284-6826

This publication is intended for guidance only and may be impacted by changes in legislation, rules, policies, and procedures adopted after the date of publication. Although this publication makes every effort to teach users how to meet applicable compliance obligations, use of this publication does not constitute the rendering of legal advice.

Michigan's Environmental Justice Policy promotes the fair, non-discriminatory treatment and meaningful involvement of Michigan's residents regarding the development, implementation, and enforcement of environmental laws, regulations, and policies by this state. Fair, non-discriminatory treatment intends that no group of people, including racial, ethnic, or low-income populations, will bear a disproportionately greater burden resulting from environmental laws, regulations, policies, and decision-making. Meaningful involvement of residents ensures an appropriate opportunity to participate in decisions about a proposed activity that will affect their environment and/or health.

**PUBLIC HEARING ON PART 625 MINERAL WELL FOR NON-HAZARDOUS
WASTE DISPOSAL**

CADILLAC SENIOR HIGH SCHOOL AUDITORIUM

CADILLAC, MICHIGAN

WEDNESDAY, MARCH 21, 2018, 6:00 – 9:00 P.M.

JENNIFER FERRIGAN - INTRODUCTION

GOOD EVENING LADIES AND GENTLEMEN, MY NAME IS JENNIFER FERRIGAN. I AM A GEOLOGIST AND PERMIT COORDINATOR AT MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY - OIL, GAS, AND MINERALS DIVISION. NEXT TO ME IS SUSANNE BITEMAN, CADILLAC DISTRICT AREA GEOLOGIST, ALSO WITH THE OIL, GAS, AND MINERALS DIVISION.

WITH US TONIGHT ARE OTHER DEQ STAFF WHO WILL BE ASSISTING WITH TONIGHT'S PROGRAM:

FROM OIL, GAS, AND MINERALS DIVISION, THIS INCLUDES:

1. RICK HENDERSON, FIELD OPERATIONS SECTION CHIEF
2. ANDY STEMCKY, CADILLAC DISTRICT SUPERVISOR
3. BOB VERSICAL, CADILLAC DISTRICT AREA GEOLOGIST
4. BILL DULEY, CADILLAC DISTRICT AREA GEOLOGIST

5. MARK SNOW, PERMITS AND BONDING UNIT SUPERVISOR

6. RAY VUGRINOVICH, MINERAL WELL SPECIALIST

AND FROM THE WASTE MANAGEMENT AND RADIOLOGICAL PROTECTION
DIVISION:

1. PHIL ROYCRAFT, CADILLAC DISTRICT SUPERVISOR

2. JIM STALEY, CADILLAC DISTRICT ENVIRONMENTAL ENGINEER

Michigan Department of Environmental
Quality
Oil, Gas, Minerals Division
(DEQ-OGMD)

Part 625 Mineral Well
Permit Application to Drill and Operate
Non-Hazardous Disposal Well

Wexford Water Technologies, LLC
Well #1 (WWT 1)



Part 625 Mineral Well
Permit Application Review Process

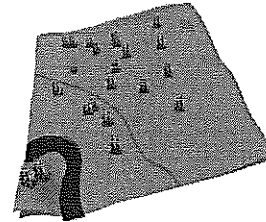
- Application Received
- Application and Site Reviewed by DEQ-OGMD Team:
 - Permit Coordinator
 - Area Geologist
 - District Supervisor
 - Other OGMD and DEQ Staff as needed
- Revisions
- Complete Application
- Public Notice of Application
- Public Comment
- Permit Decision



Part 625 Mineral Well Permit Application Review Process

- **Surface Site Location & Impacted Areas**

- Drill Permitting: Surface Location Pad, Access, Flowlines
- Land Use, Topography
- Soil Erosion & Sedimentation Control
- Ownership, Zoning



- **Environmental Impact Assessment**

- Isolation Distances
- Structures, Water Wells, Utilities
- Surface Water, Wetlands, Flood Plains
- Threatened & Endangered Species, Historic Areas
- Environmentally Sensitive Areas



Part 625 Mineral Well Permit Application Review Process

- Reserve Pit/Tanks for Cuttings
- Blow Out Preventer (BOP)
 - Meet Location Specific Criteria
 - Local Geologic Hazards and Expected Pressures
- Casing & Sealing
 - Surface Casing
 - Groundwater Protection
 - 100' into Competent Bedrock and below lowest known freshwater zone
- Sealing: Types and Volumes of Cement



Part 625 Mineral Well Permit Application Review Process

Disposal Well Specific

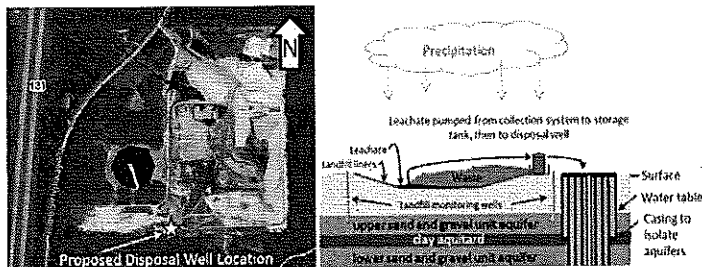
- Area of Review of Injection Interval
 - ¼ mile to 2 miles for Non-Hazardous Waste Disposal Wells
 - Plugging records of abandoned wells
 - Completion records of producing wells
- Geology
 - Local and regional scale
 - Geologic Maps & Cross Sections
 - Aquifer Review
 - Injection and Confining Zone Information
- Waste Characterization
 - Analytical
 - Waste Analysis Plan



WWT 1 Application Review

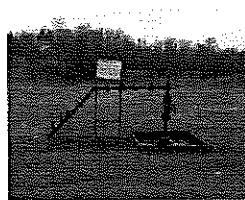
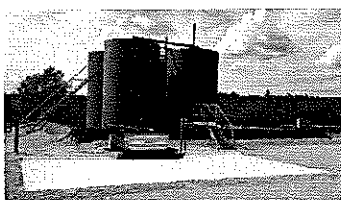
Reviewed permit application and groundwater data

- Location move to avoid any existing contamination
- Extra string of well casing to isolate upper aquifer
- Protection to prevent any carry down during the drilling process



Secondary Containment and Surface Facilities

- Secondary Containment required at wellheads, tank batteries, transfer pumps, etc.
 - Constructed in and sealed in a manner to prevent the seepage of stored liquid into soil and groundwater
- Applicant must submit plans for surface facility that include secondary containment, flow line maps, etc. to the DEQ for review and approval prior to construction.
- Groundwater monitoring system
 - Monitoring well downgradient of secondary containment
 - Hydrogeological study - baseline water chemistry right at the surface facility location
 - Or tertiary containment monitoring



Operation Inspections

- Field Staff Activities
 - Approval of Secondary Containment Plans
 - Flow Line Pressure Testing & Groundwater Monitoring
 - Response & Remediation of Spills
 - Cleanup to Part 201 Criteria
- Routine Inspections
 - Part 625 Compliance
 - Check Injection Tubing Pressure and Annulus Pressures
 - Look for Leaks, Drips, etc.
- Soil Erosion Sedimentation Control
- Response to Inquiries and Complaints
- Plugging and Restoration
 - Ensure that all oil, gas, brine, and fresh water is confined to the strata in which the oil, gas, brine, and fresh water occur by using cement plugs



JENNIFER FERRIGAN - HEARING STATEMENT

THE PURPOSE OF DEQ PARTICIPATING IN TONIGHT'S MEETING IS TO ANSWER QUESTIONS AND RECEIVE PUBLIC COMMENT ON WEXFORD WATER TECHNOLOGIES PERMIT APPLICATION TO DRILL AND OPERATE A NON-HAZARDOUS WASTE DISPOSAL WELL.

PLEASE NOTE THAT THE DEQ PUBLIC COMMENT PERIOD ENDS ON WEDNESDAY, MARCH 28, 2018, AT 5PM. A COPY OF THE PERMIT APPLICATION CAN BE FOUND ON OUR WEBSITE AT WWW.MICHIGAN.GOV/DEQLAND. THE MINERAL WELLS TAB HAS INFORMATION ON THIS APPLICATION AND A LINK TO DOWNLOAD THE APPLICATION.

PLEASE SUBMIT ANY ADDITIONAL COMMENTS IN WRITING TO:

DEQ-OGMDPUBLICCOMMENTS@MICHIGAN.GOV OR BY US MAIL TO:

MDEQ-OGMD

RE: COMMENTS WEXFORD WATER TECHNOLOGIES APPLICATION

PO BOX 30256

LANSING, MICHIGAN 48909

INFORMATION ON HOW TO SUBMIT A WRITTEN COMMENT CAN ALSO BE FOUND ON THE DEQ HANDOUT.

DEQ WILL CONSIDER ALL COMMENTS MADE, BOTH SPOKEN TONIGHT, AND SUBMITTED AS A WRITTEN COMMENT. THESE COMMENTS WILL BE CONSIDERED RELATIVE TO THE REGULATORY STANDARDS OF PART 625, MINERAL WELLS, WHEN MAKING A DECISION ON THIS PERMIT APPLICATION.

IF DURING THE HEARING PORTION OF TONIGHT'S MEETING, YOU HAVE A QUESTION, SOME OF THE DEQ STAFF WE INTRODUCED WILL BE AVAILABLE OUTSIDE THE AUDITORIUM.

THANK YOU

In Ferrigan's notes

Henry 3-21-18

Q & A

- Are Reservations Available Online

- What is ch?

- how

Question for WWT, EPA, DED

- ~~IS~~ Cana

How long have we had this type of
disposal
liability

23 Questions

- flowlines

How much

Injecting with pressure

What is injectate

• Fracking waste injection well systems

If applicant

Background check

TX 2003... It's not

business area

well behind boyfriend's home well 13

Can they stop it?

LLC... what assets?

why

\$33,000

Construction

Sniff that juice... take max rent
a

Antrim Shale

List of Contaminates

800 = 50 states
wells

who here drinks cedar creek public water
supply

Jeff Gustafson
→ handles EPA

Flowlines -

Potential for time limited

400 injection wells for EPA in Michigan

Public Comment

37/13 min

3-21-18

Michael MacCreedy

Judy Nichols - Haring Township / Cedar Creek
Garry Brown - elected official - Mayor of Monticello

Mike Solomon - West Dean Commission

No SESCP for the landfill?

One B requires A for the well.

Marcus Riccia - (CA) Mgr of City of Kadoka

Beverly Monroe - doesn't like first surveying
wants A deeper

Mike Smith - lives in Northville, Ann Arbor formerly

Jeff Dietlin - UHARCO Director for City of

Cadillac - no modifications fear of leachate
outside

Larry Pagnetti - neighbor of landfill

Daniel Stetten - 1/2 mile from "dump"

John Fuscare - protection of residents

Nancy Scungio - talking about her neighbors
she family, wants to decide to vote

Wendy Johnson - Cedar Creek resident

Shari Anstett - reclass to HAZ?

Gary Gremel - resident of Cedar Creek, alternatives
pipeline to WWT

Steve Riwowski - ant of \$ for bonding, becomes
future citizens' problem

Rita McNamara - when does problem start?

John Bennett - 12,000 pounds limit

Christopher Crain -
1 Below

Patrick

Ron Dextram - GTH

Karen Turnbull - MWC opposes permit & primary

Wendy Nystrom - MWC wants an injection
well & monitorium, this deep out of the ground

Jeff Ostrowski - MWC - treat pollution,
not bury - Deny.

Diane McGuckin - passed

Chuck Walker

Frank McNabb Cedar Creek Twp

worried about increased pressures

& fracking

Mary Hallett - Twp Treasurer Cedar Creek

Adam Coleman - Community doesn't want it,

repositives

Ferrigan, Jennifer (DEQ)

From: Greenhagen, Andrew <Greenhagen.Andrew@epa.gov>
Sent: Thursday, March 22, 2018 3:28 PM
To: Biteman, Susanne (DEQ); Ferrigan, Jennifer (DEQ)
Subject: Wexford Sign-in List
Attachments: 20180322140915935.pdf

Hi-

Attached is a scan of the sign-in list from yesterday. Please let me know if something isn't legible and we can try to interpret from the originals....if the handwriting allows.

Thanks,
Andrew

Andrew Greenhagen
Underground Injection Control Branch
U.S. EPA - Region 5
(312) 353-7648

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Name?

Well Name: Well No. 1

Permittee Name: Wexford Water Technologies

Permit #: MI-165-11-0002

	Comment at hearing?	First Name	Last Name	Mailing Address OR Email
Ex. Yes		John	Doe	123 Main, City, ST 12345
Ex. No		Jane	Smith	janesmith2@email.com
✓ * 1	Yes	Michael	MacCready	MichaelMacCready@yahoo.com
2	No	Daniel	Valley	valleydaniel-r@gmail.com
3	No	SAUN	Treiber	jetreiber@hotmail.com
4	NO	Gregory	FENTON	gm.fenton249663@yahoo.com
✓ * 5	Yes	Beverly	Monroe	monroeachers@yahoo.com
✓ * 6	Yes	Marcus	Peccia	mpeccia@cadillac-mi.net
7	No	Dan	Schling	1275 W. Long Lk 49601 - Cadillac
8	No	Wally	SMITH	1350 EL. LK R.D MAIL
9	No	Julie	Theobald	926 Sundberg st Cadillac, MI 49601
10	no	Kevin	Reinhardt	2561 W 41 rd Manton MI

⑥

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Ex. Yes		John	Doe	123 Main, City, ST 12345
Ex. No		Jane	Smith	janesmith2@email.com
1	Ro	Kirk/Leslie	Saile	lesliemethyle@charter.net
2	no	C Kim	Polzin	3771 N 45 1/2 Rd - Manton 49663
3	NO	Virginia	Bier	10329 E. 44 1/2 Rd, Cadillac 49601
✓ * 4	Yes	mike	Smith	16921 ALBONQUIN NORTHVILLE MI 48168
✓ * 5	Yes	mike	Selmon	173 S. Davidson Cadillac
6	NO	BILL	SCHNITKER	3130 N. 37 RD. MANTON 49663
✓ * 7	Yes	Jeff	DETZIN	UTILITIES@CADILLAC-MI.NET
✓ * 8	yes	LARRY	PAQUETTE	9889 E 22RD MANTON MI 49663
9	NO	JERRY	Root	1881 45th MANTON MI 49663
✓ * 10	Yes	Daniel	Sutton	TKG tkgdano@gmail.com

④

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Ex. Yes		John	Doe	123 Main, City, ST 12345
Ex. No		Jane	Smith	janesmith2@email.com
1	NO	FRED	McNABB	586-604-5969
✓ * 2	YES	JOHN	FUSCONE	2590 N. 41 RD MANTON
3	No	Patti	FUSCONE	2590 N. 41 RD MANTON
4	No	Michael	Johnson	5351 E. 20 1/2 Mile Rd. Manton
5	No	Paul	PENNER	9371 EAST 22 RD MANTON
✓ * 6	yes	Nancy	Scungio	2287 N. 41 1/2 Rd Manton
7	No	Barb	Hager	#170 Box 252 Manton
* 8		Dave	Newhouse	dnewhouse@gmail.com
9	NO	Val	Johnson	5836 N 43 Rd. Manton MI 49663
✓ * 10	yes	Judy	Nichols	1147 S 37 Rd. Canfield MI 49601

③

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Permit #: MI-165-11-0002

Permittee Name: Wexford Water Technologies

	Comment at hearing?	First Name	Last Name	Mailing Address OR Email
Ex. Yes		John	Doe	123 Main, City, ST 12345
Ex. No		Jane	Smith	jan smith2@email.com
✓ * 1	Yes	Wendy	Johnson	Jerngson@yahoo.com
✓ * 2	Yes	Rick	Anstett	9340 E 18 1/2 Rd Manton 49663
✓ * 3	Yes	Sharon	Anstett	" " " "
4	No	Dianne	Newman	PO Box 354 MANTON MI 49663
5	No	Roger	NEWMAN	" " " " " "
6	No	Mary	Newhouse	newhousemdg@charter.net 49661
7	No	Timothy	Meyer	8287 Valley Forge Dr Cadillac MI
8	No	Chad	Dylgert	8709 E 20 1/2 Rd Manton MI 49663
9	No	Alex	GIFTOS	acgiftos@yahoo.com
10	No	Geoff	Harrison	geoffreydharrison@gmail.com

⑥

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Ex. ☒	Yes	John	Doe	123 Main, City, ST 12345
Ex. ☐	No	Jane	Smith	janesmith2@email.com
✓ * 1	YES	GARY	Gremel	8603 E 20 1/2 Rd Manton
2	No	MILLER	SAUL	3567 W-LAN Mitchell Dr Cadillac MI 49601
✓ * 3	YES	GARRY	BROWN	MAYOR Manton - 232 Roberts ST Manton MI 49663
✓ * 4	YES	SEVE	PIWOWARSKI	STEVE PIWOWARSKI 55 @ GMAIL.COM
5	NO	JAN	YORKMAN	71 E 32 Rd Boon, MI 49618
✓ * 6	YES	RITA	MCNAMARA	1361 N. 4 1/2 Rd Manton, MI 49663
✓ * 7	YES	JON	Bennett	jonathanedwardbennett@gmail.com
8	NO	Edie Caldwell		9028 E. 30 1/2 Rd Cadillac MI 49601
X * 9	YES	CHARLES	BIGELOW	1220 N 21 1/2 Rd Boon MI 49618
10	NO	Elizabeth	Shamus	1310 N US 131 Manton 49663

217
3
51

⑤

Well Name: Well No. 1
Permit #: MI-145-11-0002

Permittee Name: Wexford Water Technologies

	Comment at hearing?	First Name	Last Name	Mailing Address OR Email
Ex. Yes		John	Doe	123 Main, City, ST 12345
Ex. No		Jane	Smith	janesmith2@email.com
1	NO	Thomas	Beller	104 S. Maple St Manton, MI 49663
2	NO	Aaron	JEFFERS	10450 E M 42 Manton, MI, 49663
3	No	Teresa	Baller	104 S. Maple Manton, MI 49663
4	NO	Bret	Bigelow	8571 E. 12 Rd Manton, MI, 49663
5	NO	Ruthann	French	119 Marble St Cadillac 49601
✓ * 6	YES	Christopher	Cruin	10240 E. 22 Rd Manton, MI 49663
X * 7	YES	SHARYN	Blackford	11950 W. Burns Rd, Manton, MI 49663
X * 8	Yes	Dennis	Patrick	3368 N Brown Rd Manton 49663
✓ * 9	Yes	Kim	Boyle	3356 N 39 1/2 Rd Manton
✓ * 10	Yes	Ron	Daxstrom	1729 N US 131, Manton

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Ex. No		Jane	Smith	janesmith2@email.com
1	No	Vicki	Garen	vickig@americanwaste.org
✓ * 2	Yes	Karen	Turnbull	Kturnbull72@gmail.com
X * 3	Yes	JoAnn	Porter	2879 N. US Hwy 131 Manton
✓ * 4	yes	Wendy	Nystrom	wendysbookcase@gmail.com
✓ * 5	yes	JEFF	Ostrowski	BARDOFFED@Hotmail.com
X * 6	Yes	Ken	Boioseanu	1981 Carter Rochester Hills MI 48306
X * 7	Yes	David	Bunce	427 Crippon St Cadillac 49601
X * 8	yes	Diane	McGuckin	1811 N. 45 RD. MANTON 49663
9	No	MARK	McGuckin	" " "
10				

4

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Ex. No		Jane	Smith	janesmith2@email.com
X * 1	yes	Chuck	Valkner	8702 E 14 RD. MANTON 49663
2	NO	David	ENSING	4100 N. Brown Rd. MANTON MI. 49663
3	NO	Richard	ENSING	2745 Brook Lane MANTON MI 49663
4	NO	Mike	Neumann	2619 N. 41 RD MANTON MI 49663
5	NO	JACK	Powell	255 W. LAKE DR. CADILLAC MI 49601
X * 6	Yes	SKIP	Goetner	2189 LS 131 S.
7	NO	Aaron	Groves	2351 N. 41 1/2 RD MANTON MI 49663
X * 8	Yes	Bob	Lohman	Manton MI 49663 Lohman Robert 1645 N. US 131 49663 @Yahoo.com
X * 9	yes	Eleana	Drunko	10129 E 1234 RD MANTON MI 49663
10	NO	Ron	Yorkman	91 E. 32 RD. BOON MI. 49618